Computing LCG from a Sequence of Numbers **PAL**

A linear congruential generator (LCG) is a triple of integers (A, C, M) used to generate pseudorandom numbers by the formula $x_{n+1} = (Ax_n + C) \bmod M$ applied recurrently to a chosen initial value x_0.
The generated sequence has a maximum period of length M if and only if the following conditions are met:

- GCD(C, M) = 1,
- $A - 1$ is divisible by each prime factor of M,
- if 4 divides M then also 4 divides $A - 1$.

The task

We are given a sequence of numbers x_1, ..., x_N generated consecutively by an unknown full period length LCG (A, C, M). We also know that M is the square of the product of several distinct primes, where each of these primes is from the range [5, $P_{\max}$]. It means $M = (p_1 * p_2 * \ldots * p_F)^2$ with distinct primes p_i not less than 5 and not greater than $P_{\max}$. The number of these distinct primes, i.e., factors of M, is unknown. In addition, it holds $1 \le A < M$, $1 \le C < M$, and $M \le M_{\max}$ for some known bound $M_{\max}$.
Our goal is to determine the triple (A, C, M).

Hint

It might be helpful to know how to solve equations of the form $ax \bmod b = c$ for an unknown x and positive integers a, b, c. By the assumption GCD(a, b) = 1, there is a unique solution in the range [0, $b-1$], namely $x = c\bar{a}_b \bmod b$, where $\bar{a}_b$ is the so-called *modular multiplicative inverse* of a with respect to the modulus b, fulfilling $a\bar{a}_b \bmod b = 1$. It can be computed by the extended Euclidean algorithm (its Java code is included below).
For example, to solve the equation $4x \bmod 9 = 5$, we first find the modular multiplicative inverse of 4, which is 7. Then we calculate $x = 5*7 \bmod 9 = 8$.

```java
public long multiplicativeInverse(long a, long modulus) {
    long s = 0, r = modulus, old_s = 1, old_r = a;
    while (r != 0) {
        long quotient = old_r / r;
        long temp = r;
        r = old_r - quotient * r;
        old_r = temp;
        temp = s;
        s = old_s - quotient * s;
        old_s = temp;
    }
    return old_s < 0 ? old_s + modulus : old_s;
}
```

Input

The input consists of two lines. The first line contains integers $P_{\max}$, $M_{\max}$, N. Here, $P_{\max}$ is the upper bound on factors of M, $M_{\max}$ is the upper bound on M, and N is the length of a pseudorandom number sequence generated by the unknown LCG. The second line contains the N pseudorandom numbers in the order in which they were generated. These numbers are separated by spaces.
The following limits hold:

$5 \le P_{\max} \le 503$,
$5^2 \le M_{\max} \le 2\times 10^{18}$,
$10 \le N \le 30$.
The data ensure there is exactly one solution.

Output

The output consists of one line containing integers A, C, M that are parameters of the unknown LCG. These integers are separated by spaces.

Example 1

Input

```
11 250 10
10 14 8 17 16 5 9 3 12 11
```

Output

```
11 4 25
```

Example 2

Input

```
13 287 10
31 48 30 26 36 11 0 3 20 2
```

Output

```
22 3 49
```

Example 3

Input

```
31 105305 10
24787 37631 1646 44850 39225 48780 9506 49421 40507 46773
```

Output

```
7338 194 64009
```

Observe that $64009 = 11^2 23^2$.

Example 4

Input

```
23 1016255 15
25862 130986 69020 136414 36718 66382 77181 69115 42184 144613 79952 144651 42260 69229 77333
```

Output

```
45816 19 148225
```

Observe that $148225 = 5^2 7^2 11^2$.

Public data

The public data set is intended for easier debugging and approximate program correctness checking. The public data set is stored also in the upload system and each time a student submits a solution it is run on the public dataset and the program output to stdout and stderr is available to him/her.

Link to public data set